

Input-Affine Conditioning of Selective State-Space Models Is Gauge-Absorbable: A Diagnostic and a Design Rule

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Paper under double-blind review

Abstract

Selective state-space models (Mamba) have reshaped long-context sequence modeling, yet which forms of conditioning *survive* their training dynamics is poorly understood—and getting it wrong is a quiet, recurring failure for practitioners. We study FiLM-style regime conditioning of a Mamba backbone—a per-block, channel-wise affine on each block’s input, the natural drop-in way to steer the input-dependent selective-scan parameters $\Delta, \mathbf{B}, \mathbf{C}$ —and find it a **decisive null** with an **actionable** mechanistic explanation. Under joint training the modulation collapses to identity and cannot be forced active even by direct supervision. The reason is mechanical: an input-side affine is a **gauge** direction that folds losslessly into the block’s own $\Delta/\mathbf{B}/\mathbf{C}$ projections, so the optimizer slides it back to identity, with a dataset- and architecture-invariant signature ($\tau \approx 7000$ steps, $R^2 > 0.999$) across FI-2010 equities, Kaggle crypto spot books, and Polymarket prediction markets. An output-side placement sharpens rather than escapes the mechanism—its scale folds into the block’s *other* (bias-free output) projection and the one non-absorbable component, an additive shift, decays in lockstep—so the null is **overdetermined**. This yields actions a practitioner can take *today* and a target for tomorrow. The immediately usable parts are negative and diagnostic: do not expect input- or output-affine regime conditioning of a selective scan to persist, and—before crediting any conditioning—verify the gate is active with a **load-bearing activation diagnostic** (an inert gate, $\text{film}_g \approx 0.004\text{--}0.010$ with reg_H at $\log R$, reported beside a positive generalization gap means the gap is an *artifact*, not a conditioning effect). The **design rule** points further out: surviving conditioning must modulate $\Delta/\mathbf{B}/\mathbf{C}$ *inside* the fused kernel—a target we validate at toy reference-scan scale, but one that needs a kernel-engineering project to deploy, not a drop-in a practitioner can ship today. We package the usable parts as a lift-and-use protocol, demonstrate the result on a financial substrate with an architecture-agnosticism check (no backbone dominates), and report a deflated economic coda for completeness.

1 Introduction

Conditioning a network on side information—a task ID, a context vector, an inferred regime—is among the most common things a practitioner does, and FiLM (a channel-wise affine $\gamma \odot x + \beta$) (Perez et al., 2018) is the default mechanism. When the backbone is a **selective state-space model** (Mamba (Gu & Dao, 2024)), there is an especially tempting place to inject it. Mamba’s scan parameters $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are *input-dependent* linear projections of each block’s input, so an affine on that input flows straight into the dynamics—and, crucially, it needs **no change to the fused CUDA/TileLang kernel**. This is the drop-in form everyone reaches for first. We show it does not work, explain mechanistically why, and turn that explanation into two things a practitioner can use: a diagnostic that catches a specific false-positive, and a design rule that says where a surviving modulation has to live.

The finding. We introduce a *RegimeFiLM* modulator that infers a categorical latent regime and emits per-block affine parameters (γ, β) , zero-initialized so the modulated backbone is *exactly* the unmodulated baseline at step 0 (an exact control). Despite the strongest activation pressure we could apply—forced-active initialization plus direct supervision of the router—the modulation decays to identity on every objective and dataset we test. The cause is a **gauge symmetry**: a constant channel-wise scale on the block input folds *exactly* into the host projections $W_\Delta / W_B / W_C$ as a right-multiplied diagonal $W_\bullet \text{diag}(\gamma)$, and the shift folds into their biases, with **no change to the function the block computes**. Identity is therefore a flat, lossless valley the optimizer slides along; joint training returns $(\gamma, \beta) \rightarrow (\mathbf{1}, \mathbf{0})$. The decay constant is dataset-invariant ($\tau \approx 7000$, $R^2 > 0.999$) precisely because the absorption direction is a property of the *block parameterization*, not the data. Moving the gate to the block *output* does not escape this—the scale folds into the bias-free output projection instead—so the null is overdetermined by gauge absorption (every direction the host weights can represent) plus reconstruction dominance (the one direction they cannot, the non-absorbable shift, which simply earns nothing).

Why this is actionable. Two upshots change what a practitioner does.

- *A diagnostic that catches a false-positive.* The activation diagnostics $\text{film}_g = \text{mean } |\gamma - 1|$ and the router entropy reg_H (uniform maximum $\log R$) are **load-bearing**. In our data a generalization-gap metric reads *positive*—with the “right” sign—while FiLM is provably inert and the treatment is *absolutely worse* in both regimes; the gap is a distributional (high – low) artifact whose sign even flips with head calibration. A practitioner who reports “conditioning improved the out-of-regime gap” without checking that the gate is actually *on* can ship an artifact.
- *A design rule that says where to put the gate.* If you want regime conditioning of a selective scan to *survive* training, the input-affine and output-affine forms will be folded back to identity. The surviving mechanism must act **between the block’s bounding projections**—i.e., modulate $\Delta / \mathbf{B} / \mathbf{C}$ inside the fused kernel—or the conditioning should not be attempted in affine form at all. We delineate the boundary not as a parameterization that survives, but as the *location* (inside the kernel) a surviving modulation would have to occupy.

Demonstration substrate. To stress the mechanism on genuinely different data we build a Mamba-3 world model for limit-order-book (LOB) microstructure in the DreamerV3/Drama lineage (Hafner et al., 2023; Wang et al., 2024; Lahoti et al., 2026), exercised on FI-2010 equities, Kaggle crypto spot books, and Polymarket prediction markets (§4). The backbone is a substrate, not a claim: under matched parameters *no* backbone dominates across datasets (§5.3)—exactly what a result about how selective scans *train*, rather than which one is best, should predict. The same study yields a deflated economic coda (§6): conditioning the *dynamics* did not help, but a decision rule over the model’s outputs is a survivable, naive-beating edge that a logistic regression on the same features reproduces—i.e., not attributable to the architecture. It is not the contribution.

2 Related Work

Selective state-space models and Mamba. Structured state-space models (S4, S5, Mamba) give $O(n)$ -complexity alternatives to Transformers (Gu et al., 2022; Smith et al., 2023; Gu & Dao, 2024). Mamba introduced *selective* state spaces in which Δ and $\mathbf{B}$ are input-dependent, enabling content-aware filtering; Mamba-2 (Dao & Gu, 2024) formalized the duality with structured masked attention, and Mamba-3 (Lahoti et al., 2026) adds multi-input/multi-output (MIMO) transitions. The property we exploit—that $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are learned linear projections of the block input—is exactly what makes an input-affine a gauge direction. We are not aware of prior work characterizing the training-time fate of FiLM-style conditioning of a selective scan.

FiLM and feature-wise conditioning. FiLM (Perez et al., 2018) injects context as a channel-wise affine and is the default conditioning primitive across vision, multi-task, and regime-

embedding settings (Lim et al., 2021). Our result is about *where* in a selective-scan block an affine can be placed and still survive joint training—a gauge constraint specific to the input-dependent projection structure, orthogonal to FiLM’s expressive-power story.

Interpretability of training dynamics and weight symmetries. A line of interpretability work studies *how* solutions form rather than *what* a fixed network computes, including reparameterization and gauge symmetries that leave the function unchanged while reshaping the loss landscape (Dinh et al., 2017). That a channel-wise affine before a linear map is redundant with that map is a known theme in the normalization literature: van Laarhoven (2017) shows weights feeding a normalization layer are scale-invariant, so L_2 regularization acts on their *scale*—and thus the effective learning rate—rather than as capacity control, and Hoffer et al. (2018) and Li & Arora (2020) re-derive weight decay as an effective-learning-rate schedule on the same symmetry. Our null is an instance of this redundancy in a setting it has not been analyzed: the absorbed affine is not a static normalization gain but a per-regime, *input-dependent* FiLM that folds into the selective scan’s own $W_\Delta/W_B/W_C$ projections. What is new is (i) the input-dependent selective-scan instantiation; (ii) the dataset- and architecture-invariant decay signature ($\tau \approx 7000$, $R^2 > 0.999$) attributing the collapse to the optimizer; (iii) the output-side placement showing the null is *overdetermined* (gauge absorption plus reconstruction dominance); and (iv) the practical consequence that the only kernel-friendly, drop-in form is exactly the absorbable one—so the redundancy is the default a practitioner reaches for, not a corner case. The actionable contribution is to turn that symmetry into a measurable diagnostic (film_g , reg_H) and a placement rule.

LOB models and regime-switching (substrate). DeepLOB (Zhang et al., 2019) and successors are discriminative LOB models on FI-2010 (Ntakaris et al., 2018); generative LOB work (Nagy et al., 2023; Backhouse et al., 2025) targets simulation fidelity; classical regime-switching identifies discrete states from price processes (Hamilton, 1989). We use these only as the demonstration substrate and benchmark context, not as baselines for the conditioning result.

3 Setup: FiLM Conditioning of a Selective Scan

Selective SSM. A discretized state-space layer maps $x_t \in \mathbb{R}^d$ to y_t via a latent state h_t : $h_t = \mathbf{A}h_{t-1} + \mathbf{B}x_t$, $y_t = \mathbf{C}h_t$. Mamba makes the parameters **input-dependent**:

$$\Delta_t = \sigma(W_\Delta x_t), \quad \mathbf{B}_t = W_B x_t, \quad \mathbf{C}_t = W_C x_t, \quad (1)$$

so any affine transformation of x_t propagates directly into all three scan parameters. This is the selection mechanism we attempt to steer.

RegimeFiLM modulator. A regime head produces a soft categorical $\alpha_t = \text{softmax}(W_r x_t^{(0)}) \in \mathbb{R}^R$ ($R = 8$) and a regime embedding $r_t = \alpha_t^\top E$; a hypernetwork maps r_t to per-block affine parameters, bounded for bf16 stability:

$$\gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma}^{(\ell)}) \in (0, 2), \quad \beta^{(\ell)} = \tanh(\tilde{\beta}^{(\ell)}) \in (-1, 1). \quad (2)$$

The hypernetwork is **zero-initialized**, so at step 0 $\gamma^{(\ell)} = \mathbf{1}$, $\beta^{(\ell)} = \mathbf{0}$ for all ℓ and the modulated backbone is *exactly* the unmodulated baseline—a clean, exact paired control. A Switch-Transformer load-balancing term (Fedus et al., 2022) maximizes the entropy of the *batch-averaged* regime distribution, $\mathcal{L}_{\text{regime}} = \lambda_{\text{ent}}(\log R - H[\tilde{\alpha}])$, regularizing marginal usage while leaving per-step assignments free to be peaky.

Two application sites. The **input-affine** site routes the regime into the scan through the selection mechanism:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \text{Mamba3}(\gamma^{(\ell)} \odot \text{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}). \quad (3)$$

As a control we also expose a **post-scan** site that gates the block output:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \gamma^{(\ell)} \odot \text{Mamba3}(\text{RMSNorm}(x_t^{(\ell-1)})) + \beta^{(\ell)}. \quad (4)$$

Both share the identical modulator; only the site differs. This isolates the gauge structure (§5): the input-side scale folds into $W_\Delta/W_B/W_C$; the post-scan scale folds into the bias-free output projection W_{out} ; only the post-scan additive shift is non-absorbable.

Diagnostics. Activation metrics gate every claim: $\text{film}_g = \text{mean } |\gamma - 1|$ (deviation of the multiplicative gate from identity), its additive counterpart $\text{film}_b = \text{mean } |\beta|$ (reported for the output-side shift in §5.2), and the router entropy reg_H (uniform maximum $\log R$, with R the regime count; $\log 4$ on FI-2010, $\log 8$ on Polymarket). **A positive generalization gap with an inert FiLM is an artifact, not a regime-conditioning effect**—the central methodological point.

4 Demonstration Substrate (condensed)

To exercise the mechanism on genuinely different distributions we instantiate the modulated backbone inside a Mamba-3 world model for LOB microstructure (DreamerV3/Drama lineage (Hafner et al., 2023; Wang et al., 2024)): a Transformer-over-depth encoder feeds a 16×16 categorical posterior and an $L=4$ stack of Mamba-3 SISO blocks ($d_{\text{state}}=128$), trained with the standard DreamerV3 loss (reconstruction + dynamics/representation KL) plus light auxiliary heads. The rest of the machinery (episodic memory, a Hawkes head, the imagination loop, the EV-head internals) is in an accompanying codebase; no conditioning result depends on any of it.

Regime axis. Beyond realized volatility we use a **predictability** axis on the underlying spot series—Kaufman efficiency ratio $\text{ER} = |s_n - s_0| / \sum_t |s_t - s_{t-1}|$ (near 1 a clean trend, near 0 a random wander), permutation entropy, and Hurst exponent—both to split the held-out evaluation into forecastable vs. random-walk regimes and, when the router collapses, to *supervise it directly*. “Is this window forecastable” is the most defensible regime signal we have; a router that will not encode even this axis is the strongest form of the null.

Datasets. (i) Polymarket binary-outcome crypto LOBs (BTC/ETH/SOL up/down at 5/15/60-min intervals, with the underlying Chainlink/Binance spot path); (ii) the public FI-2010 benchmark (Ntakaris et al., 2018); (iii) Kaggle high-frequency crypto spot books (BTC/ETH/ADA at 1 s / 1 min / 5 min). Held-out markets are split by predictability (primary) and realized volatility (secondary), both reported. **All thresholds are fit on the training split and frozen before any held-out evaluation**, and baseline and treatment share the seed (an exact paired control).

5 The Null and Its Mechanism

5.1 RegimeFiLM is a decisive null

Table 1 reports the central diagnostic across five objectives spanning two datasets and a money objective. In every family FiLM collapses to identity under joint training: film_g decays toward zero and reg_H sits at its uniform maximum ($\log R$). Where a generalization gap reads positive (direction +0.008, settlement +0.18, by sign), FiLM is provably inert, the treatment is *absolutely worse* in both regimes, and the gap is a high – low distributional artifact whose sign even flips with head calibration—which is exactly why the activation diagnostics are load-bearing. We flag this against ourselves: read *without* the gate diagnostic, both gaps carry the sign a successful conditioner would produce, and we would have reported a spurious win—it is only the inert film_g (with $\text{reg}_H = \log R$) that exposes the gap as an artifact. The false-positive is not a strawman; it is the reading we ourselves nearly published.

Strongest-null test. Forcing FiLM active (zero-init scale raised so $\text{film}_g = 0.150$ at step 0) and supervising the router *directly* on the spot efficiency-ratio axis with the strongest settings (LR multiplier 50, supervision weight 30, decoupled router/FiLM, fed observation volatility) still decays the modulation monotonically toward identity (film_g : $0.150 \rightarrow 0.119 \rightarrow 0.100 \rightarrow$

Family	Objective	Dataset	Gap (sign, unit)	film _g	Verdict
Volatility/scale	StudentT NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Order-flow volume	Volume NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Direction	macro-F1	FI-2010	$+0.008^+$ F1	0.018	NULL
Settlement	BCE log-loss	Polymarket	$+0.18^+$ nats	0.008	NULL
PnL	simulated trade	Polymarket	$-\$627^+$	0.010	NULL

Table 1: RegimeFiLM is inert across every objective. reg_H equals its uniform maximum ($\log R$) in all rows. The “Gap” column is in each objective’s own units (reconstruction NLL nats, macro-F1 points, BCE log-loss nats, dollars), so its magnitude is not comparable across rows; only its sign and the inert film_g are. [†]a non-zero gap whose sign is a distributional artifact, not a FiLM effect (FiLM is inert, and the treatment is absolutely worse). The PnL row’s $-\$627$ is the FiLM A/B gap on the PnL *objective* (treatment–control), *not* an economic total (cf. the $+\$5,147$ settlement total in §6). The direction-row $\text{film}_g = 0.018$ is the endpoint of the *supervised* escalation on that head (the joint-trained value is ≈ 0.004 , cf. Table 2); the settlement (0.008) and PnL (0.010) rows are joint-trained—all are inert, but the column mixes the two measurement types.

Dataset	Asset / resolution	Gap (pred / vol)	film _g	Verdict
FI-2010	event stream	$\approx 0 / \approx 0$	0.004	NULL
Kaggle BTC	1 min	$\approx 0 / \approx 0$	0.004	NULL
Kaggle ETH	1 min	≈ 0	0.005	NULL
Kaggle ADA	1 min	≈ 0	0.005	NULL
Kaggle BTC	5 min	≈ 0	0.005	NULL
Kaggle BTC	1 s (slice)	≈ 0	0.005	NULL

Table 2: Cross-dataset confirmation of the regime-FiLM null. $\text{reg}_H = \log 4$ (uniform router) in every row. The null is dataset-, asset-, and resolution-independent.

185 0.074 $\rightarrow$ 0.056 $\rightarrow$ 0.043, still falling) with $\text{reg}_H = \log 8$ throughout. The capacity is present
186 but unused: a *frozen* router probes to 0.92 agreement on the *volatility* bucket—the axis
187 the representation encodes best—yet even there the modulation collapses, while on the
188 *predictability* axis this escalation supervises the regime decodes only weakly (≈ 0.25 –0.36).

189 5.2 The mechanism: gauge absorption, and why it is overdetermined

190 **Cross-dataset invariance points at the optimizer.** We take the architecture finding to
191 FI-2010 (re-examined under the predictability-primary axis) and the Kaggle crypto spot
192 book (BTC/ETH/ADA at three resolutions), judged by forecasting metrics alone. The
193 zero-initialized modulation never leaves identity: $\text{film}_g \approx 0.004$ and $\text{reg}_H = \log 4$ in every
194 setting (Table 2). With FiLM an exact paired control, every held-out gap is statistically
195 indistinguishable from zero (all 95% CIs include 0) on both regime axes. The split is
196 non-degenerate: high-efficiency-ratio reference windows carry genuinely more structure
197 on all three predictability measures, and DeepLOB attains 0.628 macro-F1 on the FI-2010
198 horizon-10 labels—so the null is a negative on *forecastable* data, not a no-signal artifact.

199 **The collapse is optimizer-driven.** The predictability-supervised strongest-null test (forced-
200 active FiLM, router supervised directly on the efficiency-ratio bucket), applied at the **input-**
201 **side** gate, decays film_g monotonically at every logged step on *both* datasets (input-side:
202 0.250 $\rightarrow$ 0.144 Kaggle, 0.250 $\rightarrow$ 0.147 FI-2010), with $\text{reg}_H \approx \log 4$ throughout *even though the*
203 *load-balance term is disabled*—so the uniform router is not a load-balance artifact. An expo-
204 nential fit $a e^{-t/\tau} + c$ matches both trajectories at $R^2 > 0.999$ with near-identical parameters
205 ($a \approx 0.30$, $\tau \approx 7000$ steps). The decay is monotone with a negative slope sustained to the
206 last logged step (endpoint $\text{film}_g \approx 0.15$, still falling), and the fitted asymptote lies at or
207 below identity ($c \lesssim -0.05$, only more negative when the lower bound on c is relaxed), so
208 the modulation extrapolates toward identity rather than to a positive plateau—we rest the

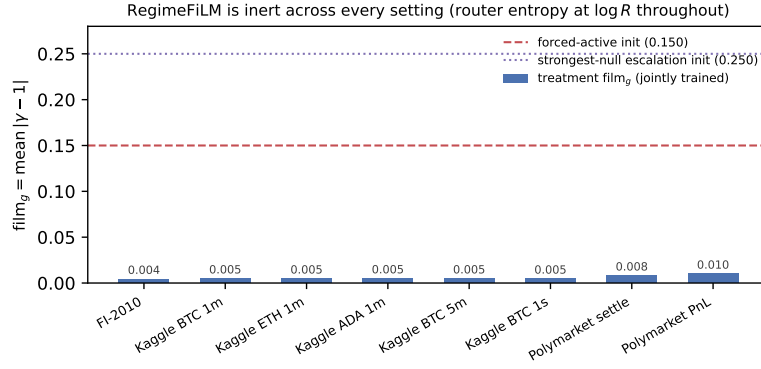


Figure 1: **RegimeFiLM is inert across every A/B setting**—the gap-positive-but-gate-off diagnostic this paper is built on. The jointly-trained treatment $\text{film}_g = \text{mean } |\gamma - 1|$ (bars) sits near 0.004–0.010 on every dataset, asset, and resolution—FI-2010, Kaggle BTC/ETH/ADA at three bar widths, and the two Polymarket objectives—while a forced-active initialization starts at 0.150 and the strongest-null escalation at 0.250 (dashed lines). The router entropy sits at its uniform maximum $\log R$ throughout. The modulation never grows from its exact identity initialization.

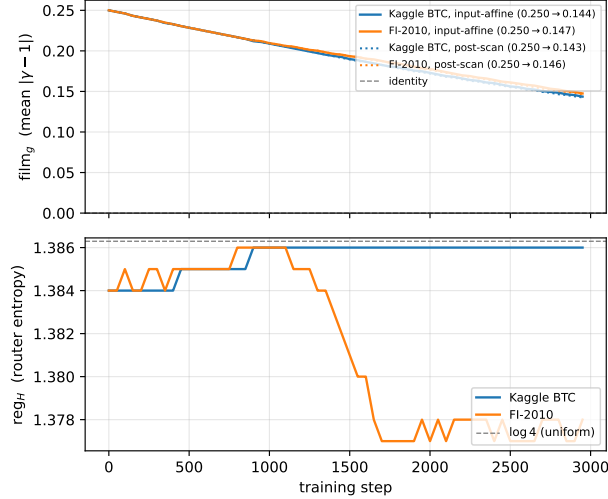


Figure 2: **Strongest-null test, and its placement-invariance.** Forcing the modulation active ($\text{film}_g = 0.250$ at step 0) and supervising the router directly on the efficiency-ratio bucket still decays film_g monotonically toward identity on both datasets (top, solid = *input-side* gate), with reg_H pinned at its uniform maximum $\log 4$ (bottom). An exponential fit gives $R^2 > 0.999$ with a dataset-invariant time constant $\tau \approx 7000$ and a negative fitted asymptote, so the modulation extrapolates toward identity rather than to a positive plateau. The *output-side* post-scan gate (top, dotted) decays along the same trajectory—its scale folds into the bias-free output projection just as the input-side scale folds into $W_\Delta / W_B / W_C$ —so the null is invariant to where the affine is applied. Endpoints differ only at the round-off level by placement (input-side 0.250 $\rightarrow$ 0.147 FI-2010 / 0.144 Kaggle; output-side 0.250 $\rightarrow$ 0.147 FI-2010 / 0.143 Kaggle, all at the last logged step).

209 claim on the observed monotone decay, with identity as the extrapolated limit. That the
 210 decay rate is essentially *identical* across two very different markets points at the optimizer,
 211 not the data (Figure 2).

Why identity is a free direction: gauge absorption. Our modulation is an input-side affine, $\tilde{x} = \gamma \odot \hat{x} + \beta$, applied to a block input that the very next operations re-project: $\Delta_t = \sigma(W_\Delta \tilde{x}_t)$, $\mathbf{B}_t = W_B \tilde{x}_t$, $\mathbf{C}_t = W_C \tilde{x}_t$. A constant channel-wise scale folds *exactly* into those projections as a right-multiplied diagonal, $W_\bullet \text{diag}(\gamma)$, and the shift folds into their biases, with **no change to the function the block computes**; the host weights can thus represent any constant (γ, β) themselves. Identity is therefore a **gauge** direction—a flat, lossless valley the optimizer slides along—so joint training folds the modulation into the host weights and returns $(\gamma, \beta) \rightarrow (1, 0)$, which is precisely the monotone decay we observe. This explains the dataset-invariant decay constant (the absorption direction is a property of the block parameterization, not the data) and *scopes* the claim: **input-affine conditioning of a selective scan is gauge-absorbable and will not persist under joint training**—not that regime conditioning is impossible in principle. The fold is exact for the *constant* part of γ ; the trained modulator is input-dependent, but on a weakly-decodable regime the supervised router collapses to uniform, γ reduces to its constant part, and weight decay folds that to identity—so the live mechanism reduces to the gauge one precisely when the regime is hard to decode (the LM control below confirms the converse). The actionable negative is that the most kernel-friendly, drop-in form of the idea is exactly the form the optimizer can and does undo.

An output-side placement: the null is overdetermined. The gauge argument names an obvious next test—gating *after* the scan—and it does *not* escape absorption. Each Mamba-3 block ends in a **bias-free** output projection $y = W_{\text{out}}h$, so a channel-wise output scale folds exactly into it, $\gamma \odot (W_{\text{out}}h) = (\text{diag}(\gamma)W_{\text{out}})h$ —the same gauge move on the block’s *other* face. The additive shift β is different: with no output bias and a nonlinear RMSNorm separating it from the next block, it is genuinely **non-absorbable**. Under the identical forced-active escalation, the output-side film_g and film_b decay in lockstep on both datasets (output-side film_g : $0.250 \rightarrow 0.147$ FI-2010, $0.250 \rightarrow 0.143$ Kaggle), with the *same* signature ($\tau \approx 7065$ FI-2010 / 6598 Kaggle, $R^2 > 0.999$). The input- and output-side endpoints, read at the same last logged step (FI-2010 0.147 vs. 0.147; Kaggle 0.144 vs. 0.143), differ only at round-off—placement changes *nothing*, which is the point. The null is therefore **overdetermined**—by two routes sharing a single driver, weight decay over a loss that rewards neither: **gauge absorption** disposes of every direction the host weights can represent (the scale on *either* face, hence the placement-invariant constant), and **reconstruction dominance** disposes of the one they cannot (the non-absorbable shift, killed because it earns nothing). A modulation escaping both would have to act *between* the block’s bounding projections—modulating $W_\Delta/W_B/W_C$ inside the fused kernel.

The boundary, demonstrated—and a gauge-vs-regularization control. Two cheap experiments turn the design rule from asserted to demonstrated. (i) *Folding*. On a toy reference scan, an input-affine modulation folds into the host projections to floating-point precision (fold residual 1.2×10^{-6} , the smallest deviation left after re-fitting the host projections to absorb the forced-active gate) and decays to identity (film_g : $0.086 \rightarrow 0.010$), while the *same* modulation applied to $\Delta/\mathbf{B}/\mathbf{C}$ *inside* the scan is non-absorbable (1.3×10^{-2} , $\sim 10^4 \times$ larger) and **persists** ($0.093 \rightarrow 0.175$, seeds 0–3)—so the in-kernel target the rule names is real. (ii) *Weight-decay control*. Removing weight decay flattens the decay (τ from ≈ 6685 to $\approx 3 \times 10^5$, R^2 $0.9999 \rightarrow 0.48$), so the gauge-direction decay is driven by **weight decay over a flat loss**, not the objective—and it is absent for the in-kernel placement.

When conditioning *does* survive: regime decodability. The collapse is not universal, and the boundary is itself actionable. A Mamba char-LM control on a two-domain corpus (English prose vs. Python source), with the regime inferred by a context-aware router supervised on the domain and the *same* input-affine FiLM, does **not** collapse: with a strongly-decodable regime (router ~ 0.95 accuracy) film_g *rises* from 0.11 to 0.52. Gauge absorption always removes the *constant* part of the affine; the *input-dependent* part survives only when the regime is decodable enough to be worth encoding—in the LOB settings the regime is weakly decodable, the router collapses to uniform, and the modulation reduces to its absorbable constant part. **The actionable rule, refined:** skip the input-affine FiLM exactly when the regime is weakly decodable from the block input; if it is strongly and

Backbone	Params	FI-2010 recon NLL	Kaggle recon NLL
Mamba-1	11.45M	−0.5403	0.0241
Mamba-2	10.20M	−0.5309	−0.0213
Mamba-3 SISO	10.30M	−0.5346	0.0207
Transformer (post-norm)	9.90M	−0.5175	0.0191
Transformer (pre-norm)	9.61M	−0.5041	0.0224

Table 3: Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). No backbone wins both datasets, so the choice of sequence model is not the operative variable.

contextually decodable, the affine *can* engage. (This control is a single-seed toy experiment; the router⇒persistence outcome is clear, but the plateau value 0.52 is not load-bearing.)

The gap is real, and FiLM closes none of it. The null is not an absence of anything to fix: on the Kaggle book the high-volatility windows are +0.105 worse in reconstruction NLL than the low-volatility windows in *both* arms, yet RegimeFiLM narrows the degradation by only ~ 0.001 while film_g is inert—it fails to engage a gap plainly present (linear-probe localization in Appendix B).

5.3 Architecture-agnosticism check (matched-parameter backbone ablation)

We compare Mamba-1, Mamba-2, Mamba-3 (SISO), and a Transformer-over-time under matched parameter budgets ($\sim 10.3\text{M}$) on FI-2010 and the Kaggle BTC spot book, reporting held-out full-channel Student- t reconstruction NLL (lower is better). The result is a genuine ambiguity: **no backbone leads on both datasets**—Mamba-1 best on FI-2010, Mamba-2 (fewest SSM params) best on Kaggle, Mamba-3 SISO competitive but not strictly dominant. A pre-norm Transformer matched to the Mamba-3 block (RMSNorm, SiLU, $4\times$ FFN; 9.61M) is likewise competitive but non-dominant (−0.5041 FI-2010, 0.0224 Kaggle), so the non-dominance is not an artifact of an under-tuned post-norm baseline. This is exactly what a result about how selective scans *train*, rather than which one is best, should predict.

Why SISO. We run Mamba-3 in SISO throughout: the MIMO kernel’s dynamic shared-memory footprint exceeds the available GPU cap at every warp-valid (rank, chunk) configuration, so the MIMO variant is outside this paper’s hardware envelope and we make no claim about it. This is the same kernel boundary the design rule names—the surviving, in-kernel modulation of $\Delta/\mathbf{B}/\mathbf{C}$ lives behind the same fused-kernel wall.

6 Deflated Economic Coda (substrate, one section)

The substrate yields a positive result the architecture did *not* carry, reported honestly as a deflated coda. Given the underlying spot path that **defines** Polymarket settlement, the world model’s directional settlement probability—traded by a selective-participation strategy that bets only when a train-frozen predictability gate fires—is a survivable, naive-beating edge whose high-predictability slice a dependence-respecting market-block bootstrap (re-sampling whole markets) places away from zero (95% CI [$+\$2,445$, $+\$7,747$] over 127 markets, $P(\text{profit}) \approx 1.0$). **Crucially the edge is architecture-independent:** a logistic regression on the *same* spot-conditioned features, traded through the identical gate and engine, reproduces and *exceeds* it ($+\$6,914$ over 243 markets vs. the settlement head’s $+\$5,147$). Binary settlement is a near-deterministic function of the spot move since open, so a linear readout captures it at least as clearly as the world model—the operative variable is the spot-conditioned pipeline and the gate, **not the sequence model**. This is the null’s lesson from the other side: conditioning the *dynamics* did not help; a decision rule over the model’s outputs did. The settlement head’s marginal low-predictability regime ($+\$51$, $P = 0.63$), its tradeable-EV-head fix, the other denominators, and the recorded-vintage reliability diagnostic are in Appendix A; the coda is not the contribution.

7 Discussion: What a Practitioner Should Do Differently

The interpretability finding is small and exact—an input-affine on a Mamba block input is a gauge direction—but it has two consequences a practitioner can act on immediately, which we package as the lift-and-use protocol below.

Diagnostic protocol: will my selective-scan conditioning survive training?

1. Instrument the gate. Log $\text{film}_g = \text{mean} |\gamma - 1|$ (deviation of the multiplicative gate from identity) and the router entropy reg_H (uniform max $\log R$) at every checkpoint. *Before* claiming any conditioning “helped,” require film_g clearly > 0 and reg_H off its uniform maximum. A positive generalization gap measured against an inert gate ($\text{film}_g \approx 0$, $\text{reg}_H = \log R$) is a distributional artifact whose sign can flip with calibration—not a conditioning effect.

2. Run the fold-residual placement test. On a tiny reference scan, fit the host projections to absorb a forced-active (γ, β) . Input- and output-affine placements fold to round-off ($\sim 10^{-6}$); a placement can *survive* only if its fold residual is large ($\sim 10^{-2}$), i.e. it acts on $\Delta/\mathbf{B}/\mathbf{C}$ *between* the block’s bounding projections (inside the fused kernel).

3. Check regime decodability. Affine conditioning engages only when the regime is strongly decodable from the block input (router accuracy high, reg_H low). If the router collapses to uniform, only the gauge-absorbable constant remains—skip the affine.

Actionable takeaway. In one line: input- and output-affine FiLM of a Mamba selective scan are gauge directions the optimizer folds back into the host $\Delta/\mathbf{B}/\mathbf{C}$ (or bias-free output) projections and decays to identity ($\tau \approx 7000$, invariant across datasets and placement)—so modulate $\Delta/\mathbf{B}/\mathbf{C}$ inside the fused kernel, or do not bother, and verify the gate is on before reporting that conditioning “helped.” The two-number check (film_g , reg_H) is cheap, architecture-agnostic, and catches a seductive false-positive—an ablation that “turns conditioning on” and reports a better out-of-regime gap that moved for distributional reasons while the conditioner never engaged.

The design rule names a target, not a dead end. What remains after gauge absorption and reconstruction dominance—modulating the scan parameters *between* the block’s bounding projections, inside the fused kernel—is the location a surviving modulation must occupy, and §5.2 confirms at toy scale that a modulation placed there *does* survive. That is a kernel-engineering task (needing the shared-memory headroom the SISO/MIMO boundary in §5.3 also runs into), and the natural next experiment.

Limitations. The decay is established at $\tau \approx 7000$ over our training horizon; the fitted asymptote lies at or below identity (robust to relaxing the lower bound on c), so the modulation extrapolates *through* identity, though the within-horizon endpoint is still slightly above it—we do not claim a positive near-identity plateau is impossible under every optimizer/schedule. The constructive claims rest on single-seed toys (the LM control and the escalation fits; the in-kernel toy is seeds 0–3), while the null is multi-setting and multi-seed. The economic coda is a backtest substrate, not part of the interpretability claim.

8 Conclusion

An input-affine FiLM on a Mamba block input is a **gauge direction** the optimizer folds back into the host projections, so it silently decays to identity ($\tau \approx 7000$, $R^2 > 0.999$, invariant across datasets and placement); the null is overdetermined by gauge absorption plus reconstruction dominance. What a practitioner can act on *today* is negative and diagnostic: do not ship input- or output-affine regime conditioning of a selective scan, and verify the gate is on (the two-number $\text{film}_g/\text{reg}_H$ check) before crediting it. The constructive **design rule**—modulate $\Delta/\mathbf{B}/\mathbf{C}$ inside the fused kernel—names the location a surviving modulation must occupy; we validate it at toy scale and leave it as the next kernel-engineering step.

340 A Deflated economic coda: figures

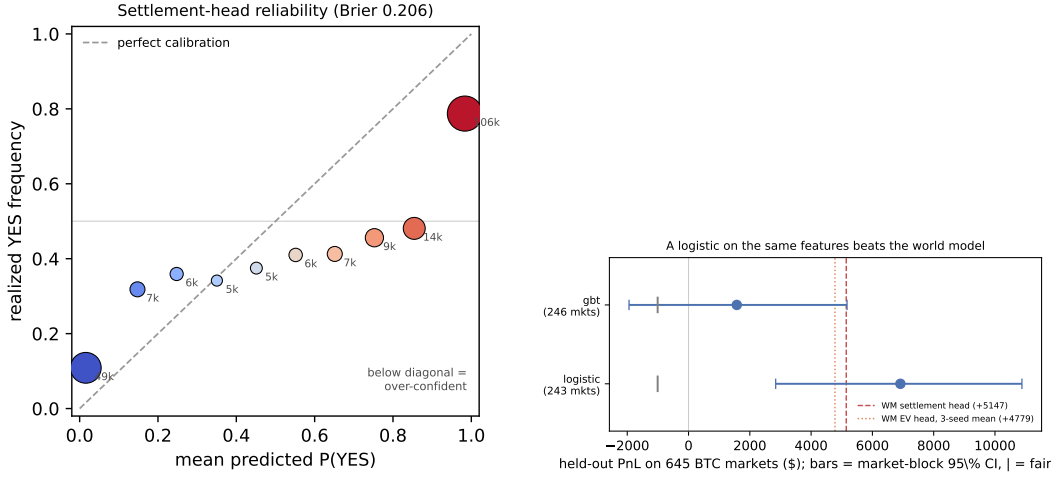


Figure 3: **Left:** settlement-head reliability over 214k held-out probabilities (Brier 0.206). Points fall below the diagonal for mean > 0.5 (systematic over-confidence), yet the populous extreme deciles land on the correct side of 0.5 (0.787 top decile, 0.109 bottom decile; 0.785/0.108 under 0.1-wide bins)—the directional signal the divergence trade harvests; mid-confidence bins (0.5–0.9) realize below 0.5, the calibration footprint of the low-predictability failure the EV head removes. **Right:** the economic edge does not need the architecture—held-out PnL of the same-feature simple models on the 645-market set, with market-block 95% CIs. The logistic’s +\$6,914 exceeds both world-model heads, and its market-block interval (+\$2,844 to +\$10,880 over 243 independent markets) survives the dependence-respecting bootstrap.

341 **Economic-coda caveats (detail).** The edge harvests the lag between the underlying spot
 342 and the contract book; walk-the-book fills assume recorded liquidity is takeable, so live
 343 market impact would erode some of it. Only the train-frozen predictability gate is *con-*
 344 *firmary*; the depth-band and early-market cuts are *exploratory*, subject to forking-paths
 345 caveats. The reliability table and PnL totals are recorded-vintage structural diagnostics—the
 346 over-confidence pattern is structural, not vintage-specific.

347 **Regime robustness via a tradeable-EV head (detail).** Supervising the *tradeable* expected
 348 value directly—a BookRelativeEdgeHead trained with Huber loss on $e_{yes} = \text{outcome} - \text{ask}_{yes}$, $e_{no} = (1 - \text{outcome}) - \text{ask}_{no}$, so the target is near zero (abstain) when the book is
 349 already fair—makes the low-predictability regime robust: low-pred +\$2,368 at $P(\text{profit}) =$
 350 0.99 (seed 0); three-seed mean +\$4,779 total at mean low-pred $P(\text{profit}) = 0.98$, collapsing
 351 the high-vs-low PnL spread $6.2\times$ (\$5,044 $\rightarrow$ \$814) at total PnL comparable to settlement
 352 (+\$5,147). The same-feature logistic is *already* regime-robust—high-pred +\$2,448, low-pred
 353 +\$4,466, both $P(\text{profit}) \geq 0.99$ —so the low-predictability signal is not intrinsically hard; the
 354 EV head is a real fix for the world-model readout, not the only way to capture it.

356 B Linear-probe localization of regime decodability

357 Two probes localize the failure behind the null. A post-hoc reading of the jointly-supervised
 358 router shows it is uniform *per step* (per-sample entropy $\approx \log 4$) with near-chance agreement
 359 against the efficiency-ratio bucket (0.25–0.31 vs. 0.25): direct supervision does not even
 360 make the router regime-discriminative. A frozen linear probe on the unmodulated features
 361 decodes the per-step predictability bucket only weakly (0.33–0.36 vs. 0.25 chance) and the
 362 volatility bucket somewhat better (0.43–0.47)—yet the volatility-axis gap is *also* null. So
 363 even on the axis the representation encodes best, and even when the regime is supplied

by supervision, the in-scan modulation collapses to identity. (The degenerate next-tick direction head is de-confounded by inverse-frequency class balancing, which yields a non-degenerate macro-F1 with the gap still null; the single borderline gap, Kaggle spot-vol at $n=2$, is disproved by a pre-registered third seed that flips its sign.)

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